

Multiple Choice (10 marks)

1. Mai earns 5.50 *per hour* *after-school* job. How many hours does she have to work to earn 132?
 - (a) 16
 - (b) 20
 - (c) 24
 - (d) 28
2. How many square units are in the region satisfying the inequalities $y \geq |x|$ and $y \leq -|x| + 3$? Express your answer as a decimal.
 - (a) 4.5
 - (b) 2.25
 - (c) 6.6
 - (d) 3.3
3. How many positive 4-digit integers have only odd digits?
 - (a) 100
 - (b) 999
 - (c) 625
 - (d) 500
4. Define a new measurement as the difference between the 60 th and 40 th percentile scores in a population. This measurement will give information concerning
 - (a) central tendency.
 - (b) variability.
 - (c) symmetry.
 - (d) skewness.
5. What is the smallest prime whose digits sum to 19?
 - (a) 5
 - (b) 11
 - (c) 17
 - (d) 199

True / False (10 marks)

Answer True or False

6. True or False: The combined width of a cardboard strip measuring 5 over 6 ft and another measuring 2 over 3 ft is 2 ft.
7. True or False: There are 8 ways to distribute 4 distinguishable balls into 2 indistinguishable boxes.
8. True or False: The solution to the equation $8y = 56$ is $y = 6$.
9. True or False: The number -1 is a solution to the equation $5x - 5 = -10$.
10. True or False: The number 11.1 is a solution to the equation $x + 2.7 = 8.4$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Consider the sequence defined by $a_k = \frac{1}{k^2+k}$ for $k \geq 1$. Given that $a_m + a_{m+1} + \cdots + a_{n-1} = \frac{1}{29}$, for positive integers m and n with $m < n$, find $m + n$.
12. Analyze the function $y = 5 + 3 * \sin(\pi - x)$ to determine its range. Explain the steps you took to arrive at your conclusion and the significance of the range in this context.

End of Paper